zetesis-puremath Blueprint

A Lean 4 pure-mathematics reservoir:
analytic measurability, information theory, concentration, minimax,
kernel embeddings, and dual VC.

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Contents

Chapter 1

Introduction

This blueprint pins every mathematical definition and headline theorem in `zetesis-puremath`[1] to its Lean 4 declaration. The companion source is a personal pure-math reservoir: content needed by the author's formalization projects (`formal-learning-theory-kernel`, SMFE, and others) that is either absent from Mathlib or specialized in a way Mathlib's current formulation does not support directly.

1.1 What is formalized here

Six substrates, each with its own chapter below:

- **Measure theory on Polish spaces.** Analytic sets are null-measurable under finite Borel measures; Kechris 30.13 capacitability theorem for analytic sets; real-valued metric total variation distance.
- **Information theory.** Real-valued Kullback-Leibler bridge to Mathlib's `ENNReal`-valued `klDiv`; binary KL with a full derivative / monotonicity chain; Pinsker's inequality with the sharp constant; mutual information and its zero-iff-product characterization.
- **Probability.** `BoundedRandomVariable` typeclass with a Chebyshev-via-majority bound; double-sample / `ValidSplit` / `SplitMeasure` exchangeability infrastructure; `FintypePMF` (an $\mathbb{R}$-valued, `linarith`- compatible specialization of Mathlib's `PMF`).
- **Decision theory.** Covering-based minimax; multiplicative-weights-update (MWU) in the Freund-Schapire $\beta = (1 - \eta)$ convention; approximate-minimax bound $T = \mathcal{O}(\log N/\varepsilon^2)$.
- **Reproducing-kernel Hilbert spaces.** Kernel mean embedding with the reproducing property; Maximum Mean Discrepancy (MMD) and characteristic kernels; Hilbert-Schmidt Independence Criterion (HSIC).
- **Combinatorics.** Assouad's 1983 dual VC bound $\mathrm{VCdim}(M^\top) \le 2^{d+1} - 1$.

1.2 Verification tiers

All 114 public theorems discharge under the following tiers, all green at commit `471e5d5` (see `test/verification/CI/`):

- **Tier 0**: `lake build`, 0 errors, 0 warnings, 0 `sorry` in `ZPM/`.

[1] https://github.com/Zetetic-Dhruv/zetesis-puremath

- **Tier 1**: `#print axioms`. 113 theorems use only `propext`, `Classical.choice`, `Quot.sound`; the 114th (`challenge_bnd_subset_cyl`) uses none.
- **Tier 2**: `leanchecker -fresh` per module (GitHub Actions `workflow_dispatch`).
- **Tier 3**: `comparator` + Landlock sandbox on GitLab CI; *"Lean default kernel accepts the solution. Your solution is okay!"* (pipeline #2463996501, duration 36 min).

Every statement in this blueprint carries a `\lean{...}` annotation linking to the Lean declaration; where the full proof is in the source, `\leanok` marks the entry as formally verified.

Chapter 2

Measure theory

2.1 Analytic measurability

Theorem 2.1 (Analytic null-measurability). *Lean:* `MeasureTheory.AnalyticSet.nullMeasurableSet`

Let α be a Polish space, μ a finite Borel measure on α, and $S \subseteq \alpha$ analytic. Then S is μ-null-measurable [14, Thm 29.5], [4, §7.4.11].

Theorem 2.2 (Inner approximation by compacts). *Lean:* `MeasureTheory.AnalyticSet.exists_isCompact_measureReal_`

Under the hypotheses of Theorem 2.1, for every $\delta > 0$ there exists a compact $K \subseteq S$ with $\mu(S) - \mu(K) < \delta$ (in $\mathbb{R}$-valued form) [14, Thm 17.11].

2.2 Choquet capacity

Definition 2.3 (Choquet capacity). Lean: `MeasureTheory.IsChoquetCapacity`

A map $I : \mathrm{Set}(\alpha) \to \mathbb{R}^{\infty}_{\geq 0}$ is a *Choquet capacity* iff (i) it is monotone, (ii) continuous from below along increasing sequences, and (iii) continuous from above along antitone sequences of closed sets [6], [14, Defn 30.1], [8, III.1]. Every finite Borel measure on a Polish space is a Choquet capacity.

Definition 2.4 (Compact capacity hull). Lean: `MeasureTheory.compactCap`

For μ and S as above,

$$\mathrm{compactCap}\,\mu\,S \ := \ \sup\{\mu(K) : K \subseteq S,\ K \text{ compact}\}.$$

Theorem 2.5 (Every finite Borel measure is a Choquet capacity). *Lean:* `MeasureTheory.measure_isChoquetCapacity`

Every finite Borel measure on a Polish space satisfies the three axioms of Definition 2.3 [8, III.7], [14, Thm 17.11].

Theorem 2.6 (Monotonicity of `compactCap`). *Lean:* `MeasureTheory.compactCap_mono`

$S \subseteq T \Rightarrow \mathrm{compactCap}\,\mu\,S \leq \mathrm{compactCap}\,\mu\,T$.

Theorem 2.7 (Measurable inner approximation). *Lean:* `MeasureTheory.MeasurableSet_compactCap_eq`

If S is measurable then $\mu(S) = \mathrm{compactCap}\,\mu\,S$ [14, Thm 17.11].

Theorem 2.8 (Kechris 30.13: analytic capacitability). *Lean:* `MeasureTheory.AnalyticSet.cap_eq_iSup_isCompact`

Let I be a Choquet capacity on a Polish space α, and S an analytic subset of α. Then

$$I(S) \ = \ \sup\{I(K) : K \subseteq S,\ K \ \textit{compact}\}$$

[14, Thm 30.13], [6], [8, III.29–III.33]. Together with Definition 2.4 and Theorem 2.5, this specializes to $\mu(S) = \mathrm{compactCap}\,\mu\,S$.

Theorem 2.9 (Analytic compact-hull equality). *Lean:* `MeasureTheory.AnalyticSet.compactCap_eq`

For analytic S, $\mu(S) = \operatorname{compactCap} \mu\, S$. *Corollary of Theorem 2.8 [14, Thm 30.13].*

2.3 Baire-space cylinder infrastructure

The proof of Theorem 2.8 requires a small theory of cylinders in $\mathbb{N}^{\mathbb{N}}$, adapted from [14, §7.A].

Definition 2.10 (Cylinder). Lean: `Cyl`

For $N : \mathbb{N} \to \mathbb{N}$ and $n \in \mathbb{N}$,

$$\operatorname{Cyl} N\, n \;:=\; \{g : \mathbb{N} \to \mathbb{N} \mid \forall i \le n,\; g\, i \le N\, i\}$$

[14, §7.A].

Definition 2.11 (Bounded box). Lean: `Bnd`

$\operatorname{Bnd} N \;:=\; \bigcap_n \operatorname{Cyl} N\, n \;=\; \prod_i [0, N_i]$ [14, §7.A].

Definition 2.12 (Truncation map). Lean: `truncate`

$\operatorname{truncate} N\, g\, i \;:=\; \min(g\, i,\; N\, i)$. Sends any sequence into $\operatorname{Bnd} N$ and agrees with g on cylinders.

Lemma 2.13 ($\operatorname{Bnd} N$ is compact). *Lean:* `isCompact_bnd`

$\operatorname{Bnd} N$ *is compact (Tychonoff applied to the finite intervals* $[0, N_i]$*).*

Lemma 2.14 ($\operatorname{Bnd} \subseteq \operatorname{Cyl}$). *Lean:* `bnd_subset_cyl`

$\operatorname{Bnd} N \subseteq \operatorname{Cyl} N\, n$ *for every* n.

Lemma 2.15 (Cylinder successor splits along the next coordinate). *Lean:* `cyl_succ_eq`

$\operatorname{Cyl} N\,(n+1) = \operatorname{Cyl} N\, n \cap \{g : g\,(n+1) \le N\,(n+1)\}$.

Lemma 2.16 (Intersection with a coordinate update). *Lean:* `cyl_inter_eq_cyl_update`

Intersecting a cylinder with a coordinate-level constraint rewrites as a cylinder with an updated bound at that coordinate.

Lemma 2.17 (Monotone split). *Lean:* `monotone_cyl_split`

The map $k \mapsto \operatorname{Cyl} N\, n \cap \{g : g\,(n+1) \le k\}$ *is monotone in* k.

Lemma 2.18 (Truncate lands in Bnd). *Lean:* `truncate_mem_bnd`

$\operatorname{truncate} N\, g \in \operatorname{Bnd} N$ *for every* g.

Lemma 2.19 (Truncate agrees on cylinders). *Lean:* `truncate_agree_on_cyl`

If $g \in \operatorname{Cyl} N\, n$ *then* $\operatorname{truncate} N\, g\, i = g\, i$ *for all* $i \le n$.

Lemma 2.20 (Cylinder extensionality). *Lean:* `cyl_ext`

Two elements of $\operatorname{Cyl} N\, n$ *agreeing on the first* $n+1$ *coordinates are equal on* Cyl.

Lemma 2.21 (Intersection of cylinder-closure images equals compact image). *Lean:* `iInter_closure_image_cyl_eq`

Under a continuous parametrization $f : \mathbb{N}^{\mathbb{N}} \to \alpha$ *into a Polish space,* $\bigcap_n \overline{f(\operatorname{Cyl} N\, n)} = f(\operatorname{Bnd} N)$.

2.4 Total variation distance

Definition 2.22 (Real-valued total variation). Lean: `MeasureTheory.tvDistReal`

For finite Borel measures P, Q on α,

$$\mathrm{TV}(P, Q) \;:=\; \sup_A \lvert P(A).\mathrm{toReal} - Q(A).\mathrm{toReal}\rvert$$

where the supremum is over measurable $A \subseteq \alpha$. This is the *half-sup* / metric convention, chosen so that Pinsker reads as $\mathrm{TV}(P, Q) \le \sqrt{\mathrm{KL}(P\|Q)/2}$ [4, §3], [24, Ch. 1].

Lemma 2.23 (Set nonempty). *Lean:* `MeasureTheory.tvDistReal_set_nonempty`

The set $\{\lvert P(A).\mathrm{toReal} - Q(A).\mathrm{toReal}\rvert : A \text{ measurable}\}$ *is nonempty (witnessed by* $A = \emptyset$*).*

Lemma 2.24 (Set bounded above). *Lean:* `MeasureTheory.tvDistReal_set_bddAbove`

The same set is bounded above by $P(\alpha).\mathrm{toReal} + Q(\alpha).\mathrm{toReal}$.

Remark 2.25. The repository also contains a second total-variation object, $\mathrm{FintypePMF.tvDistance}\, p\, q = \sum_a \lvert p\, a - q\, a\rvert$ (the L^1 convention, see Chapter 4); the two conventions differ by a factor of 2 on discrete supports. Downstream consumers should not compose them without inserting the factor-of-$\frac{1}{2}$ correction.

Chapter 3

Information theory

3.1 Real-valued KL bridge

Definition 3.1 (Real-valued KL). Lean: `InformationTheory.klDivReal`

For measures P, Q on a measurable space, set

$$\mathrm{KL}_{\mathbb{R}}(P,Q) \;:=\; \begin{cases} \int \log \frac{\mathrm{d}P}{\mathrm{d}Q}\,\mathrm{d}P & \text{if } P \ll Q, \\ 0 & \text{otherwise.} \end{cases}$$

This bridges Mathlib's `ENNReal`-valued `klDiv` to $\mathbb{R}$ for `linarith`-compatible arithmetic; the identity $\mathrm{KL}_{\mathbb{R}}(P,Q) = (\mathrm{KL}(P,Q)).\mathrm{toReal}$ holds unconditionally (`klDivReal_eq_toReal_klDiv`) [7, §2.3], [16, §2.1].

Theorem 3.2 (Non-negativity). *Lean:* `InformationTheory.klDivReal_nonneg`

$\mathrm{KL}_{\mathbb{R}}(P,Q) \geq 0$ *for all measures (Gibbs' inequality) [7, Thm 2.6.3].*

Theorem 3.3 (Zero iff equal, under finiteness). *Lean:* `InformationTheory.klDivReal_eq_zero_iff`

If $P \ll Q$*,* $\mathrm{KL}(P,Q) \neq \infty$*, and both are finite measures, then* $\mathrm{KL}_{\mathbb{R}}(P,Q) = 0 \iff P = Q$ *[7, Thm 2.6.3].*

3.2 Binary KL and Pinsker's inequality

Definition 3.4 (Binary KL). Lean: `InformationTheory.klBin`

For $p, q \in [0,1]$,

$$\mathrm{KL}_{\mathrm{bin}}(p,q) \;:=\; p \log \tfrac{p}{q} + (1-p) \log \tfrac{1-p}{1-q}$$

with the convention $0 \cdot \log(0/\cdot) = 0$ [7, §2.3, eq. (2.26)], [16, §2.1].

Lemma 3.5 (Derivative in q). *Lean:* `InformationTheory.hasDerivAt_klBin_q`

For $q \in (0,1)$*,* $\partial_q \mathrm{KL}_{\mathrm{bin}}(p,q) = (q-p)/(q(1-q))$ *[16, §2.1, eq. (2.6)], [5, §4.4].*

Theorem 3.6 (Binary Pinsker, sharp constant 2). *Lean:* `InformationTheory.binary_pinsker`

For $p \in [0,1]$ *and* $q \in (0,1)$*,* $2(p-q)^2 \;\leq\; \mathrm{KL}_{\mathrm{bin}}(p,q)$ *[15], [7, Lemma 11.6.1].*

Theorem 3.7 (Pinsker's inequality). *Lean:* `InformationTheory.pinsker_proof`

For probability measures $P \ll Q$ *with* $\mathrm{KL}(P,Q) < \infty$*,*

$$\mathrm{TV}(P,Q) \;\leq\; \sqrt{\mathrm{KL}_{\mathbb{R}}(P,Q)/2}$$

[7, Thm 11.6.1]. The convention matches Definition 2.22 (half-sup TV); with the L^1 *convention the bound reads* $\|P-Q\|_1 \leq \sqrt{2\,\mathrm{KL}}$*.*

Theorem 3.8 (Binary-to-general coarsening). *Lean:* `InformationTheory.klBin_le_klDivReal`

For every measurable A *with* $P(A) = p$, $Q(A) = q$, $\mathrm{KL}_{\mathrm{bin}}(p, q) \le \mathrm{KL}_{\mathbb{R}}(P, Q)$. *This is the two-point data-processing inequality and is the step linking the binary Pinsker to the general measure-theoretic Pinsker [7, Thm 2.8.1].*

3.3 Mutual information

Definition 3.9 (Real-valued mutual information). Lean: `InformationTheory.mutualInformationReal`

For a probability measure P on $X \times Y$ with marginals P_X, P_Y, $\mathrm{MI}_{\mathbb{R}}(P) := \mathrm{KL}_{\mathbb{R}}(P, P_X \otimes P_Y)$ [7, §2.3, Defn 2.28], [16, §3.1].

Theorem 3.10 (Non-negativity of mutual information). *Lean:* `InformationTheory.mutualInformationReal_nonneg`

$\mathrm{MI}_{\mathbb{R}}(P) \ge 0$ *(Gibbs applied to the joint vs. product-of-marginals) [7, Thm 2.6.3].*

Theorem 3.11 (Mutual information vanishes iff independent). *Lean:* `InformationTheory.mutualInformationReal_eq_`

Under $P \ll P_X \otimes P_Y$ *and* $\mathrm{KL}(P, P_X \otimes P_Y) \ne \infty$, $\mathrm{MI}_{\mathbb{R}}(P) = 0 \iff P = P_X \otimes P_Y$ *[7, Thm 2.6.5].*

Chapter 4

Probability

4.1 Bounded random variables + concentration

Definition 4.1 (Bounded random variable). Lean: `ProbabilityTheory.BoundedRandomVariable`

$f : \Omega \to \mathbb{R}$ is *bounded* in $[a, b]$ under μ iff f is measurable and $f(\omega) \in [a, b]$ for μ-almost every ω [5, §2.6].

Theorem 4.2 (Chebyshev majority bound). *Lean:* `ProbabilityTheory.chebyshev_majority_bound`

Let $X_1, \ldots, X_k$ be independent random variables with each $\mu(A_j) \geq 2/3$ where $A_j = \{X_j = \text{correct}\}$. Then for $k \geq 9/\delta$, the probability that a majority of the X_j is correct is at least $1 - \delta$. Proof via Popoviciu's variance bound + Chebyshev's inequality [5, §2].

4.2 Exchangeability and double-sample

Definition 4.3 (Sample bundle). Lean: `ProbabilityTheory.ExchangeableSample`

A bundle $\langle m, \mu, \text{IsProbabilityMeasure}\,\mu\rangle$ where $m \in \mathbb{N}$ is the sample size and μ is a probability measure on the sample space. Used with the product measure $\mu^{\otimes 2m}$ as the natural exchangeable double-sample domain [13, §11], [5, Ch. 12].

Definition 4.4 (Valid split). Lean: `ProbabilityTheory.ValidSplit`

$s : \text{Fin}(2m) \to \text{Bool}$ is a *valid split* iff exactly m of its values are `true`. There are $\binom{2m}{m}$ valid splits (Vapnik-Chervonenkis double-sample [23]).

Definition 4.5 (Uniform split measure). Lean: `ProbabilityTheory.SplitMeasure`

The uniform probability measure on the set of valid splits. This is the symmetrization measure for the double-sample trick [23], [5, Ch. 12].

4.3 Finite-type PMF

Definition 4.6 (Real-valued fintype PMF). Lean: `ProbabilityTheory.FintypePMF`

For a fintype α, a `FintypePMF` α is a pair $\langle p, h_{\geq 0}, h_{\sum = 1}\rangle$ where $p : \alpha \to \mathbb{R}$, $p \geq 0$ pointwise, and $\sum_{a:\alpha} p(a) = 1$. This is an $\mathbb{R}$-valued, `linarith`-compatible specialization of Mathlib's $\mathbb{R}^{\infty}_{\geq 0}$-valued `PMF` [5, §4].

Theorem 4.7 (Bridge to Mathlib PMF). *Lean:* `ProbabilityTheory.FintypePMF.toPMF_apply`

The map $\text{toPMF} : \text{FintypePMF}\,\alpha \to \text{PMF}\,\alpha$ *satisfies* $(\text{toPMF}\,p)(a) = \text{ENNReal.ofReal}(p.\text{prob}\,a)$, *which is lossless on valid FintypePMF inputs [5, §4.1].*

Definition 4.8 (L^1 TV distance)**.** Lean: `ProbabilityTheory.FintypePMF.tvDistance`

tvDistance $p\,q$ $:=$ $\sum_{a:\alpha} |p(a) - q(a)|$. This is the *full* L^1 convention, not the half-sup TV of Definition 2.22; on discrete supports they differ by a factor of 2 [5, §4.1], [24, Ch. 1].

Theorem 4.9 (Non-negativity of tvDistance)**.** *Lean:* `ProbabilityTheory.FintypePMF.tvDistance_nonneg`

tvDistance$(p, q) \geq 0$ *[5, §4.1].*

Theorem 4.10 (Symmetry of tvDistance)**.** *Lean:* `ProbabilityTheory.FintypePMF.tvDistance_comm`

tvDistance$(p, q) =$ tvDistance(q, p) *[5, §4.1].*

Theorem 4.11 (Self-distance is zero)**.** *Lean:* `ProbabilityTheory.FintypePMF.tvDistance_self`

tvDistance$(p, p) = 0$ *[5, §4.1].*

Theorem 4.12 (Transfer principle)**.** *Lean:* `ProbabilityTheory.FintypePMF.expectation_approx_of_tv`

For Boolean-valued f, $\left|\mathbb{E}_p[f] - \mathbb{E}_q[f]\right| \leq$ tvDistance(p, q). In the half-sup TV convention this would read $\leq 2 \cdot \mathrm{TV}(p, q)$ [5, §4.1].

Chapter 5

Decision theory: minimax and MWU

5.1 Boolean-matrix zero-sum games

Definition 5.1 (Boolean-matrix payoff). Lean: `ProbabilityTheory.boolGamePayoff`

For a Boolean matrix $M : R \to C \to \mathrm{Bool}$, a row distribution $p : \mathrm{FintypePMF}\, R$, and a pure column $c : C$,

$$\mathrm{boolGamePayoff}\, M\, p\, c \;:=\; \sum_{r:R} p(r) \cdot [M\, r\, c].$$

This is the expected $\{0, 1\}$-gain of the row player, against a pure column play, in the matrix game M [10].

Theorem 5.2 (One-sided minimax sanity). *Lean: `ProbabilityTheory.minimax_value_le_one`*

If the row player can guarantee value v against every mixed column (the `hrow` hypothesis), then $v \le 1$ [10].

5.2 Multiplicative weights update

Definition 5.3 (MWU config). Lean: `ProbabilityTheory.MWUConfig`

An `MWUConfig` C is a pair $\langle w, h_{>0}\rangle$ where $w : C \to \mathbb{R}$ is the strictly positive weight vector. The *potential* is $\Phi(w) := \sum_c w(c)$, the *induced PMF* is $\mathrm{toPMF}(w)(c) := w(c)/\Phi(w)$ [3, Thm 2.1].

Definition 5.4 (Update and closed-form). Lean: `ProbabilityTheory.mwuUpdateWeights`

Given a row $r : R$ and step size $\eta < 1$, the update is $w'(c) := w(c) \cdot (1 - \eta)$ if $M\, r\, c = \mathrm{true}$, else $w'(c) := w(c)$. This is the Freund-Schapire $\beta = (1 - \eta)$ Hedge convention [9]. The closed form

$$w_T(c) \;=\; w_0(c) \cdot (1 - \eta)^{\mathrm{hitCount}_T(c)}$$

is Theorem `mwu_weight_eq_pow_hitCount`.

Theorem 5.5 (One-step potential bound). *Lean: `ProbabilityTheory.potential_one_step_bound`*

If the row player can guarantee value v against every mixed column and $0 \le \eta < 1$, then after one MWU step $\Phi_{t+1} \le \Phi_t \cdot (1 - \eta v)$ [3, Cor. 2.2].

Theorem 5.6 (T-step potential bound). *Lean: `ProbabilityTheory.mwu_potential_T_bound`*

Induction gives $\Phi_T \;\le\; |C| \cdot (1 - \eta v)^T$ [3, Thm 2.1].

Theorem 5.7 (Closed-form weight). *Lean: `ProbabilityTheory.mwu_weight_eq_pow_hitCount`*

$w_T(c) = w_0(c) \cdot (1 - \eta)^{\mathrm{hitCount}_T(c)}$ *[9, eq. (2)].*

5.3 Approximate minimax

Theorem 5.8 (Approximate minimax via MWU). *Lean:* `ProbabilityTheory.mwu_approx_minimax`

Fix $\varepsilon > 0$, set $\eta = \varepsilon/4$ and $T = \max\{1, \lceil 16 \log N/\varepsilon^2 \rceil\}$ where $N = |C|$. If the row player can guarantee value v against every mixed column, then the empirical row mixture produced by T rounds of MWU achieves

$$v - \varepsilon \;\le\; \text{boolGamePayoff}\, M\, p\, c \qquad \textit{for every pure } c \in C.$$

This is the Freund-Schapire constructive minimax [10, Thm 3]; [3, §3.1] unifies this with general online learning.

Theorem 5.9 (Covering-based minimax existence). *Lean:* `ProbabilityTheory.covering_minimax`

Given the `hrow` *hypothesis, there exists a mixed-row PMF whose payoff against every pure column is at least $1/|C|$. Proof by pigeonhole-averaging over per-column pure-row witnesses extracted via* `ProbabilityTheory.exists_covering_row` *[10].*

Theorem 5.10 (ε-approximate finite minimax). *Lean:* `ProbabilityTheory.finite_approx_minimax`

Under a feasibility side-condition $v - \varepsilon \le 1/|C|$, the covering-based mixed-row strategy from Theorem 5.9 achieves payoff $\ge v - \varepsilon$ against every pure column. This is the logical dual of the quantitative MWU bound in Theorem 5.8 [10], [3, §3.1].

Chapter 6

Reproducing-kernel Hilbert spaces

Background conventions: the Mathlib class `RKHS H X` is the scalar-valued reproducing-kernel Hilbert space on a domain X, with Hilbert space $\mathcal{H} = H$ and feature map $k_x := \mathrm{kerFun}\, H\, x\, 1 \in H$ satisfying $f(x) = \langle k_x, f\rangle$ [1]. Throughout, $\mathcal{H}$ denotes a complete, separable, real inner product space.

6.1 Bounded kernels

Definition 6.1 (Bounded kernel)**.** Lean: `BoundedKernel`

A `BoundedKernel` $\mathcal{H}$ is a witness $\langle C, h_{\geq 0}, h_{\leq}\rangle$ with $C \geq 0$ and $\|k_x\| \leq C$ for all $x : X$. Bounded-feature kernels are the canonical setting for kernel mean embeddings [21, §4.2].

Theorem 6.2 (Evaluation bound)**.** *Lean:* `norm_apply_le_bound_mul_norm`

$|f(x)| \leq C \cdot \|f\|$ for every $f \in \mathcal{H}$ and $x \in X$ [21, Thm 4.23].

Theorem 6.3 (Reproducing identity)**.** *Lean:* `rkhs_eval_eq_inner`

$f(x) = \langle k_x,\, f\rangle$ for every $f \in \mathcal{H}$ and $x \in X$ [1].

Theorem 6.4 (Continuity of RKHS evaluations)**.** *Lean:* `continuous_rkhs_apply`

If the feature map $x \mapsto k_x$ is continuous, then every $f \in \mathcal{H}$ is continuous on X [21, Lemma 4.29].

Theorem 6.5 (RKHS functions are integrable under a finite measure)**.** *Lean:* `rkhs_fun_integrable`

Under `IsFiniteMeasure` P and a continuous feature map, every $f \in \mathcal{H}$ is P-integrable (via $L^\infty \subset L^1$ on finite measures).

6.2 Kernel mean embedding

Definition 6.6 (Kernel mean embedding)**.** Lean: `kernelMeanEmbedding`

For a measure P on X, set

$$\mu_P \;:=\; \int_X k_x \,\mathrm{d}P(x) \;\in\; \mathcal{H}$$

(Bochner integral) [19].

Theorem 6.7 (Reproducing property for the embedding)**.** *Lean:* `kernelMeanEmbedding_reproducing`

For integrable f, $\langle \mu_P, f\rangle \;=\; \int_X f(x)\,\mathrm{d}P(x)$ [19, eq. (4)], [11, §2, eq. (2)].

Theorem 6.8 (Norm bound)**.** *Lean:* `kernelMeanEmbedding_norm_le`

Under `IsProbabilityMeasure` P, $\|\mu_P\| \leq C$ [11, Lemma 4].

Theorem 6.9 (Integrability of the KME integrand). *Lean:* `kernelMeanEmbedding_integrable`

Under `IsFiniteMeasure` P*, continuity of* $x \mapsto k_x$*, and a bounded kernel, the feature map* $x \mapsto k_x$ *is Bochner-integrable against* P*, so* $\int_X k_x \, \mathrm{d}P(x)$ *is well-defined [11, §2].*

6.3 Maximum Mean Discrepancy

Definition 6.10 (Squared MMD). Lean: `mmdSq`

$\mathrm{MMD}^2(P, Q) \ := \ \|\mu_P - \mu_Q\|^2$ [11, Defn. 2].

Theorem 6.11 (Non-negativity of MMD^2). *Lean:* `mmdSq_nonneg`

$\mathrm{MMD}^2(P, Q) \geq 0$ *[11, Lemma 5].*

Theorem 6.12 (MMD^2 on identical measures). *Lean:* `mmdSq_self`

$\mathrm{MMD}^2(P, P) = 0$ *[11, Lemma 5].*

Theorem 6.13 (Symmetry of MMD^2). *Lean:* `mmdSq_comm`

$\mathrm{MMD}^2(P, Q) = \mathrm{MMD}^2(Q, P)$ *[11, Lemma 5].*

Definition 6.14 (Characteristic RKHS). Lean: `IsCharacteristic`

$\mathcal{H}$ is *characteristic on* X iff the kernel mean embedding $P \mapsto \mu_P$ is injective. This is the modern definition [20, Defn. 6].

Theorem 6.15 (Zero-iff-equal under characteristic kernel). *Lean:* `mmdSq_zero_iff`

Under `IsCharacteristic` $\mathcal{H}$, $\mathrm{MMD}^2(P, Q) = 0 \iff P = Q$ *[20, Thm 8], [11, Thm 5].*

Theorem 6.16 (Forward direction: equal measures ⇒ zero MMD). *Lean:* `mmdSq_eq_zero`

$P = Q \Rightarrow \mathrm{MMD}^2(P, Q) = 0$ *(direct consequence of* $\mu_P = \mu_Q$*) [11, Lemma 5].*

6.4 Hilbert-Schmidt Independence Criterion

Definition 6.17 (HSIC via product-space MMD). Lean: `hsicDef`

For a joint measure P on $X \times Y$ with marginals P_X, P_Y,

$$\mathrm{HSIC}(P) \ := \ \mathrm{MMD}^2(P, \ P_X \otimes P_Y).$$

This is the modern unified formulation [20, Thm 24]; originally introduced as a cross-covariance-operator Hilbert-Schmidt norm [12].

Theorem 6.18 (Non-negativity of HSIC). *Lean:* `hsicDef_nonneg`

$\mathrm{HSIC}(P) \geq 0$ *[12].*

Theorem 6.19 (HSIC vanishes on independent measures). *Lean:* `hsicDef_eq_zero_of_independent`

If $P = P_X \otimes P_Y$ *then* $\mathrm{HSIC}(P) = 0$ *(direct consequence of* $\mathrm{MMD}^2(P, P) = 0$*) [12].*

Theorem 6.20 (HSIC zero iff independent). *Lean:* `hsicDef_zero_iff_independent`

Under `IsCharacteristic` $\mathcal{H}$ *on* $X \times Y$, $\mathrm{HSIC}(P) = 0 \iff P = P_X \otimes P_Y$ *[12, Thm 4], [20, §4.2].*

Chapter 7

Combinatorics: dual VC

7.1 Binary matrix setup

Definition 7.1 (Binary matrix). Lean: `BinaryMatrix`

$\mathrm{BinaryMatrix}\, m\, n \;:=\; \mathrm{Fin}\, m \to \mathrm{Fin}\, n \to \mathrm{Bool}$. Type-definitionally equal to $\mathrm{Matrix}\,(\mathrm{Fin}\, m)\,(\mathrm{Fin}\, n)\,\mathrm{Bool}$.

Definition 7.2 (Induced set family). Lean: `BinaryMatrix.toFinsetFamily`

The *induced set family* of a binary matrix $M : \mathrm{Fin}\, m \to \mathrm{Fin}\, n \to \mathrm{Bool}$ consists of the true-supports of its rows, $\{\{j : M\, i\, j = \mathrm{true}\} : i \in \mathrm{Fin}\, m\}$ [23], [17].

Definition 7.3 (Matrix shattering). Lean: `BinaryMatrix.shatters`

For a binary matrix M and $S \subseteq \mathrm{Fin}\, n$, M *shatters* S iff every pattern $t \subseteq S$ is realized by some row of M [23], [17].

Theorem 7.4 (Consistency with `Finset.Shatters`). *Lean:* `BinaryMatrix.shatters_iff`

M shatters $S \iff \mathrm{toFinsetFamily}\, M$ shatters S in Mathlib's sense [17].

Lemma 7.5 (Boolean function equality via filter). *Lean:* `BinaryMatrix.bool_fun_eq_of_filter_eq`

Two Boolean-valued functions $f, g : \mathrm{Fin}\, n \to \mathrm{Bool}$ with equal true-filters are equal.

7.2 Sauer-Shelah corollary

Theorem 7.6 (Cardinality of the induced family). *Lean:* `BinaryMatrix.card_toFinsetFamily_le`

If $\mathrm{VCdim}(\mathrm{toFinsetFamily}\, M) \le d$ then $|\mathrm{toFinsetFamily}\, M| \le \sum_{k=0}^{d} \binom{n}{k}$. This is the Sauer-Shelah bound [17], [18], [23] routed through Mathlib's `Finset.card_shatterer_le_sum_vcDim` (Pajor + Sauer-Shelah) into the matrix setting.

7.3 Assouad's dual VC bound

Lemma 7.7 (Dual-to-primal shattering coding). *Lean:* `BinaryMatrix.transpose_shatters_imp_shatters`

If $M^\top$ shatters $S \subseteq \mathrm{Fin}\, m$ with $|S| \ge 2^{d+1}$, then M shatters some $T \subseteq \mathrm{Fin}\, n$ with $|T| = d + 1$. Proof by a bitstring-coding argument: over $(d+1)$-bit patterns on S, each pattern must be realized by a distinct column, giving $d + 1$ distinguishing columns that M shatters [2, Thm 2.13].

Theorem 7.8 (Assouad's bound: $\mathrm{VCdim}(M^\top) \le 2^{d+1}-1$). *Lean:* `BinaryMatrix.assouad_transpose_vcDim`

If $\mathrm{VCdim}(M) \le d$ then $\mathrm{VCdim}(M^\top) \le 2^{d+1} - 1$ [2, Thm 2.13]. The bound is tight at small d; no strengthening is known for general d without additional structure.

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